An Area-Efficient Lookup-Table -Based eDRAM Digital CIM Macro for Neural Network Inference

Yifan He, *Member, IEEE*, Shupei Fan, *Graduate Student Member, IEEE*, Xuan Li, *Graduate Student Member, IEEE*, Luchang Lei, *Graduate Student Member, IEEE*, Wenbin Jia, *Graduate Student Member, IEEE*, Chen Tang, *Member, IEEE*, Yaolei Li, *Graduate Student Member, IEEE*, Zongle Huang, *Graduate Student Member, IEEE*, Zhike Du, Jinshan Yue, *Member, IEEE*, Xueqing Li, *Senior Member, IEEE*, Huazhong Yang, *Fellow, IEEE*, Hongyang Jia, *Member, IEEE*, and Yongpan Liu, *Senior Member, IEEE*

***Abstract*—Digital circuit is a promising approach to implement computing-in-memory (CIM) architecture for data-intensive applications, such as neural network inference. Previous digital CIM implementations have demonstrated higher throughput and robustness than the analog counterpart, benefiting from technology scaling and full-precision datapaths. However, the traditional digital CIM designs adopt a bulky adder tree as the computing circuit, limiting further improvements in area and energy efficiency. To address the above limitation, this work proposes a novel design methodology that combines the lookup-table (LUT) and high-density embedded dynamic RAM (eDRAM) cells. By storing precomputed weight summations in LUTs, the proposed approach reduces the first two stages of the adder tree. Additionally, the row-level parallelism of the CIM architecture is utilized to amortize the overheads of LUT weight encoding and eDRAM refreshing. The 28-nm prototype demonstrates a peak area efficiency of 16.2 TOPS/mm^2 and an energy efficiency of 49.5 TOPS/W, both for 8-bit operations. The proposed eDRAM macro achieves a 2.4 and 0.48 Mb/mm^2 density in memory and CIM mode, due to its balanced computing-storage ratio, enabling reconfigurability between computing cores and memory storage.**

***Index Terms*—Computing-in-memory (CIM), digital circuit, embedded dynamic RAM (eDRAM), lookup-table (LUT).**

I. Introduction

MATRIX-VECTOR multiplication lies in the heart of modern deep neural network models, where the huge weight parameters and massive multiply-and-accumulation (MAC) operations put severe challenges for traditional computing platforms, such as CPU and GPU [1], [2]. The fundamental problem is due to the inefficient data transmission between memory array and computing units for classic Von Neumann architectures [3], [4]. Therefore, the computing-in-memory (CIM) architecture is proposed to minimize the overhead by equipping memory array with computing capability [5], [6]. Such approach enables both parallel readout and in situ dimension reduction, improving the throughput and efficiency at the same time [7], [8], [9].

The proposal of digital CIM [10], [11] marks a milestone in the history of CIM developments. The digital approach directly integrates standard logic cells aside the bit cell. Compared with previous analog implementations [8], [12], [13], [14], the standard logic cell guarantees the robustness for reliable neural network inference. Also, the digital circuits can benefit more from the advanced technology with increased transistor density and lower operating voltage, which brings both higher area and energy efficiency simultaneously [15], [16], [17]. Most importantly, the digital CIM macro provides easier integration with the digital SoC, especially when scaling up to large-scale CIM systems [18]. Actually, the developments of digital CIM have led to the widespread industrial application of CIM technology, ranging from embedded memories to DRAM [19], [20], [21].

Although previous digital CIM implementations have demonstrated the advantages of digital CIM in terms of accuracy, performance, and scalability, the efficiency is fundamentally limited by the bulky digital logic [22]. As shown in Fig. 1, a typical digital CIM macro can be transformed into a traditional weight stationary PE array by replacing the memory bit cells with latches. Considering that both adopt the same digital computing circuits, the main difference lies in the storage devices. However, the area of a fully parallel digital CIM array is mostly occupied by the massive logic cells, making the choice of memory cells less significant. As a result, previous digital CIM macros [11], [15], [16], [17], due to its limited storage density, behave more like a computing engine, instead of a memory array. This leads us to an important question, if the benefit of digital CIM solely comes from customized logic cells and manual layout?

Received 3 March 2025; revised 25 May 2025; accepted 21 July 2025. Date of publication 18 August 2025; date of current version 30 March 2026. This article was approved by Associate Editor Meng-Fan Chang. This work was supported by Beijing Major Science and Technology Project under Contract Z241100004224016. *(Corresponding author: Yongpan Liu.)*

Yifan He was with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China. He is now with the School of Electrical and Computer Engineering, Cornell Tech, New York, NY 10044 USA.

Shupei Fan, Luchang Lei, Wenbin Jia, Chen Tang, Yaolei Li, Zongle Huang, Zhike Du, Xueqing Li, Huazhong Yang, Hongyang Jia, and Yongpan Liu are with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: ypliu@tsinghua.edu.cn).

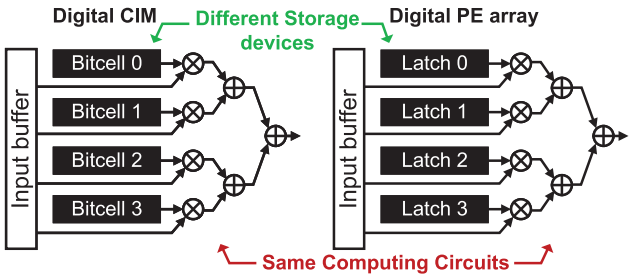
Xuan Li is with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China, and also with Wiener Technology, Beijing 100080, China.

Jinshan Yue is with the Institute of Microelectronics of the Chinese Academy of Sciences, Beijing 100029, China.

Digital Object Identifier 10.1109/JSSC.2025.3595012

0018-9200 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.



(a)

	Standard Digital CIM	Projected PE Array
Cell Area (um²)	0.379 (6T SRAM)	0.921 (16T Latch, +143%)
Capacity (Kb)	64	64
Macro Area (mm²)	0.202	0.238 (+17.8%)
Storage Density (Kb/mm²)	316.8	269.4 (-14.9%)

(b)

Fig. 1. (a) Illustration of digital CIM and traditional weight-stationary digital PE array. (b) Comparison of a 22-nm digital CIM macro [11] and the PE array projected by replacing SRAM with latches.

Sun et al. [23] summarize the mainstream digital CIM optimization approaches ranging from customized full adder to algorithm co-design. Although improving energy efficiency significantly, most approaches can also be applied to a traditional digital design flow by merging the customized cells into standard cell libraries. In other words, both the efficiency of digital CIM and digital accelerators are still fundamentally limited by digital circuits.

In this work, we break through the aforementioned limitation by revisiting the digital CIM design methodology. Rather than directly accumulating the results via full adders, we encode the weight summations into an area-efficient lookup-table (LUT) made up of high-density embedded dynamic RAM (eDRAM) cells. This allows the MAC operations to be replaced by efficient memory reads, thereby eliminating the first two stages of the adder tree. The proposed eDRAM-LUT design leverages two feature of CIM architecture simultaneously: stationary weights and high-density memory cells. This distinguishes our design from previous digital CIM work that still relied on the traditional digital circuit design methodology. The main contributions of our work are listed as follows.

1) We propose an area-efficient eDRAM-LUT-based adder (eLAD) tree, which reduces 42% area and 58% energy than a conventional digital CIM design.
2) We reduce the overhead of LUT encoding and eDRAM refreshing by 2.9× and 16× via an in-memory acceleration approach.
3) We propose a high-density computing-storage reconfigurable eDRAM CIM macro with a density FoM[1] 5.8× higher than previous designs.

This article is an extended version of [25]. Compared to the previous conference version, we split the voltage supplies of computing circuits and eDRAM write drivers to achieve lower operating voltage and 2.5× higher energy efficiency. The eDRAM cell is also optimized with an improved biasing scheme to extend the retention time by 7.7×. The layout of other computing and peripheral circuits is also optimized to reduce the whole macro area by 5.5%.

[1]The density FoM is defined as the product of storage density and 8-bit MAC operation area efficiency [24].

II. Background and Challenges

A. Design Motivations

While the LUT-based computing technique has been explored since the emergence of CNN accelerators [26], [27], it was first adopted in digital CIM by [28], where a logic-based LUT was implemented to reduce dynamic power. Compared to the conventional digital CIM design, the logic-based LUT replaces the first stage of the adder tree with a 4-to-1 multiplexer. By using a static adder to compute the weight summation, the total dynamic power is reduced by 21%. While such approach does utilize the weight stationary property of the CIM architecture, its digital logic area is actually increased due to the additional logic, which requires more input bit parallelism for amortization. The proposed eDRAM-LUT design moves one step further by building a memory-based LUT design. The area overhead of LUT can be fully tolerated by the high-density memory cells, which allows us to exploit more aggressive adder tree reduction. Compared to the both previous LUT-based CIM and traditional digital accelerators [26], [27], [29], which still rely on standard cells to build LUT, the proposed eDRAM-LUT design is both area and energy efficient for general MAC operations.

On the other hand, our eDRAM-LUT-based digital CIM architecture is also motivated by previous reconfigurable analog CIM designs [30], [31]. The capacitors from the charge-based CIM array can be reused as the CDAC in the SAR-ADC for area reduction. Similarly, the CIM macro built by high-density eDRAM LUTs can be reconfigured as a normal memory buffer to reduce the data transfer between different macros. The effectiveness of the dual-mode CIM has been demonstrated in previous work with reconfigurable dataflow [24] and optimized mapping strategy [32]. Compared to previous analog reconfigurable designs that rely on a multiplexer to switch between ADC and CIM modes, the eDRAM-LUT design enables reconfiguration by simply updating the LUT content, incurring minimal area overhead while fully utilizing all bit-cells.

B. Challenges to Efficient eDRAM-LUT-Based CIM

The key challenge to build an efficient LUT-based digital CIM macro is the high-density memory cell design. Only when the inflated weight array has a smaller area than corresponding accumulation circuits, can the LUT-based optimization bring benefits. However, previous eDRAM-based CIM work typically employed thick oxide transistors or extra capacitors at the storage node to improve the retention time. The resulted cell area, though having less transistors, was even larger than an SRAM cell at the same technology node [30], [33]. Such solutions sacrificing the storage density for refreshing overhead are incompatible with the principle of LUT-based computing. As shown in Fig. 2(a), we adopt a compact eDRAM cell layout without additional MOM capacitors. As a result, the retention time is degraded by 10×. In such case,

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

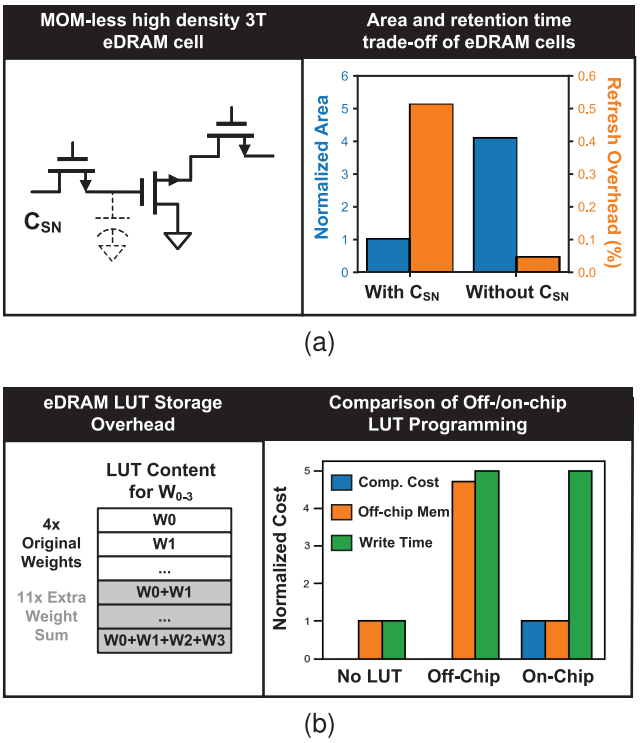
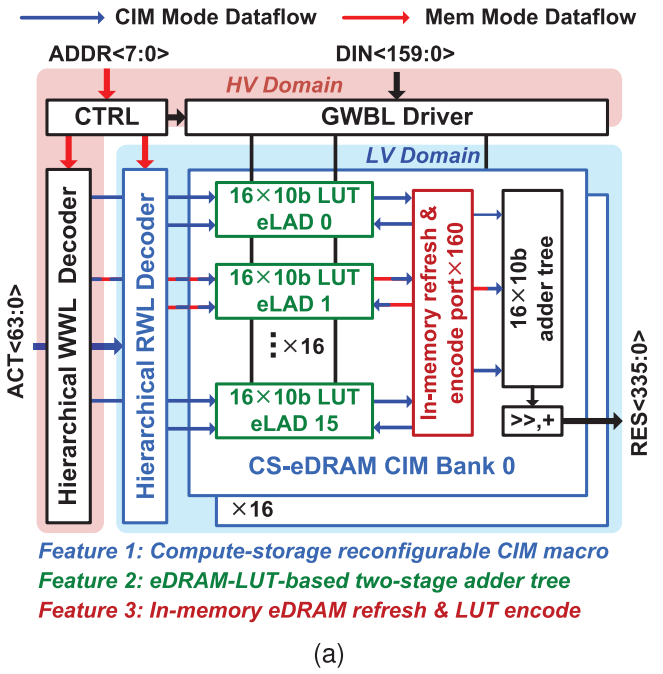


Fig. 2. Main challenges of eDRAM-LUT-based digital CIM. (a) Refreshing overhead of MOM-less high-density eDRAM and (b) LUT programming overhead.



Modules	CIM Mode	Memory Mode	Voltage Domain
Top controller	Bypassed MSB decoder	4-to-16 MSB WL decoder	HV Domain
Hierarchical WWL decoder & GWBL driver	4-16 LSB decoder×16 & GWBL buffer	4-16 LSB decoder×1 & GWBL buffer	HV Domain
Hierarchical RWL decoder	4-16 local decoder×16	4-16 local decoder×1	LV Domain
16×10b LUT	4×8b weights with 16 combinations	16×10b storage	LV Domain
Adder tree	Accumulation circuits	Readout circuits	LV Domain

(b)

Fig. 3. (a) Overall architecture of the proposed eDRAM CIM macro with three key features and (b) function mapping of main components for CIM and normal memory modes.

frequent periodic refreshing operation will incur more than 50% overhead for computation operations at the worst scenario with high temperature and low operating frequency.

Besides the eDRAM issue, the LUT itself also introduces an additional programming cost, since both original weights and their linear combination need to be stored. Fig. 2(b) compares two typical solutions: off-chip and on-chip LUT encoding. While first one avoids the weight generation overhead by precomputing the LUT content during compilation stage, its drawback is also obvious since both off-chip storage and programming time inflates by 5×. As a tradeoff between on- and off-chip resources, the on-chip encoding approach locally accumulates the weight summation with additional computation units. However, both approaches still need to load weights into CIM macro serially, which inevitably consumes more programming time limited by I/O bandwidth.

As a summary, implementing a compact memory cell and reducing the encoding cost are essential to make eDRAM-LUT-based CIM practical and efficient.

III. Reconfigurable eDRAM-LUT-Based Digital CIM Macro Architecture

The overall architecture of the proposed eDRAM-LUT-based digital CIM macro is shown in Fig. 3(a), highlighting three key innovations. First of all, the design consists of a 40-Kb reconfigurable eDRAM array that serves dual purposes for both computing and storage. Unlike the digital CIM macros relying solely on logic circuits for computation [11], [22], [34], the proposed LUT-based implementation achieves the storage density higher than a logic-rule SRAM [35] in memory mode while maintaining full input and output parallelism for high-throughput CIM operations. The whole eDRAM array is partitioned into 16 banks corresponding to different output channels, amortizing the WL decoder costs of 64-bit inputs.

The differences between CIM and normal memory mode mainly lie in the decoder configuration and LUT content. As shown in Fig. 3(b), the decoder is decomposed into LSB and MSB parts. In the memory mode, the MSB decoder is activated to select one of the 16 LSB decoders based on ADDR<7:4>, while the remaining address bits are forwarded to the selected LSB decoder to activate the corresponding WL. In contrast, the CIM mode bypasses the MSB decoder and enables all 16 LSB decoders, allowing for parallel LUT-based computation. The activation bits are directly used to fetch the corresponding partial sums. Consequently, the LUT content in CIM mode consists of the 16 combinations of 4×8 b weights instead of the 16×10 b arbitrary data in the memory mode. The detailed operation of both CIM and memory modes is illustrated later in Section IV-B.

Second, each CIM bank incorporates of 16 eLAD modules to perform efficient MAC operations. The eLAD is equivalent

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

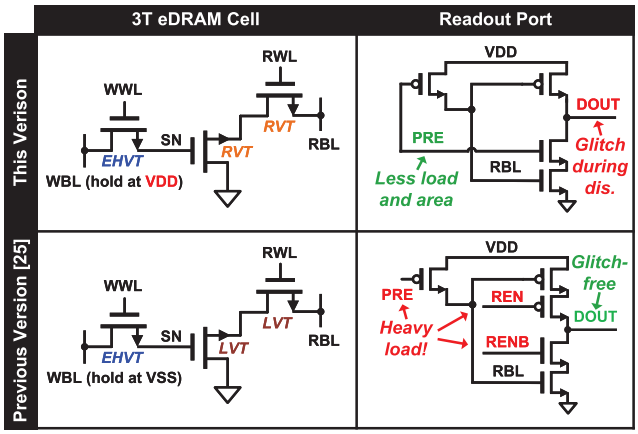


Fig. 4. Schematic of 3T eDRAM cells in this work and previous version [25] with different biasing schemes and readout port designs.

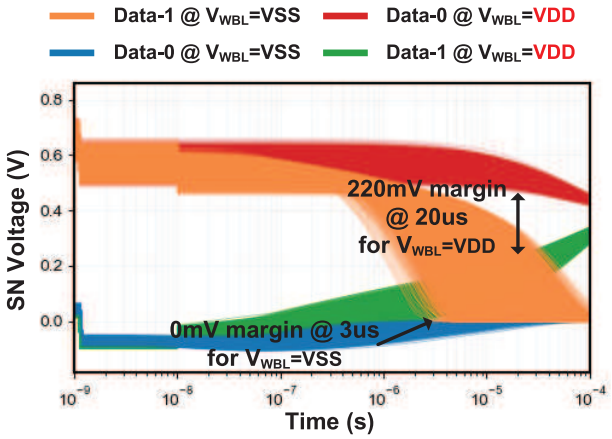
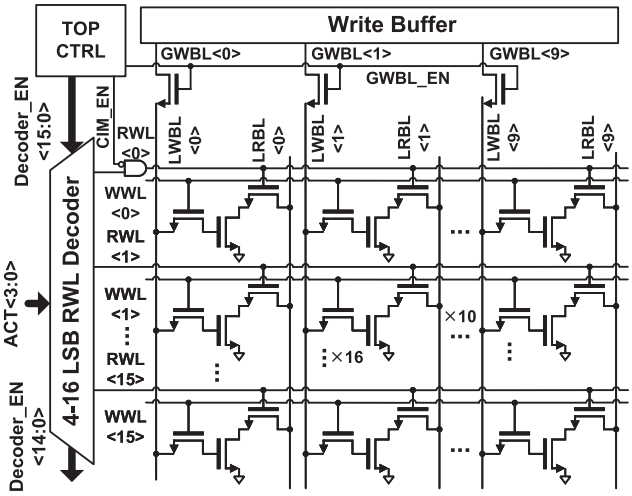


Fig. 5. 10k-point Monte Carlo simulation of leakage test at 60 °C and different WBL bias voltages.



WL	Data	WL	Data	WL	Data	WL	Data
0	Don't Care	4	W_3	8	W_4	12	W_3+W_4
1	W_1	5	W_1+W_3	9	W_1+W_4	13	$W_1+W_3+W_4$
2	W_2	6	W_2+W_3	10	W_2+W_3	14	$W_2+W_3+W_4$
3	W_1+W_2	7	$W_1+W_2+W_3$	11	$W_1+W_2+W_4$	15	$W_1+W_2+W_3+W_4$

(c)

Fig. 6. (a) Schematic of eLADs, (b) layout implementation of the 3 T eDRAM cell compared to a standard nine-track full adder cell, and (c) data mapping of eLAD in CIM mode.

to the first two stages of an adder tree, generating the MAC results of four input channels. Compared to the conventional digital logic implementations, the eLAD replaces complex full adders with simple 3 T eDRAM cells. Additionally, the dynamic power consumption caused by frequent switching is reduced through low-power memory read operations, thereby enabling both area- and energy-efficient MAC operations, as illustrated later in Section IV-B. The outputs of eLAD modules are subsequently aggregated through a 16 × 10-bit adder tree. To minimize the digital logic area, the CIM macro adopts bit-serial processing for input activations. Thus, the shift adders are employed to accumulate the partial sums in an LSB-first order over eight cycles, generating full-precision results for 64-input 8-bit MAC operations.

Lastly, dedicated local read and write ports are placed between eLAD modules and the following accumulation circuits. The bidirectional ports enable flexible and parallel data movements within the eDRAM array, an essential feature for efficient in-memory refreshing and weight encoding operations, as detailed in Section V. The macro also includes other peripherals for data movement and controlling, such as the I/O interfaces, timing modules, and the driver circuits for

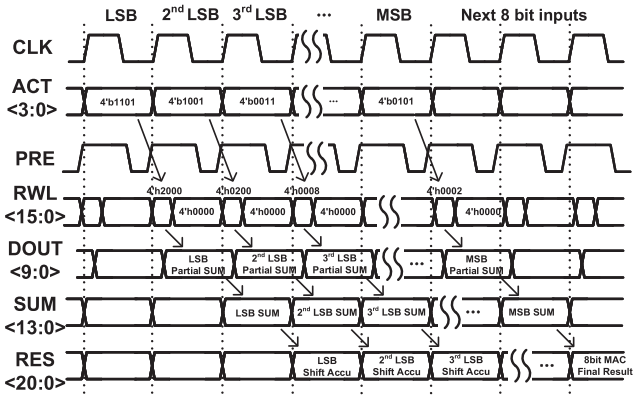


Fig. 7. Timing diagram of eLAD-based CIM operations.

global signals. Notably, different from [25], the write circuits, including WWL decoders, GWBL, and WWL drivers, are grouped in an isolated high-voltage domain, separating them

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

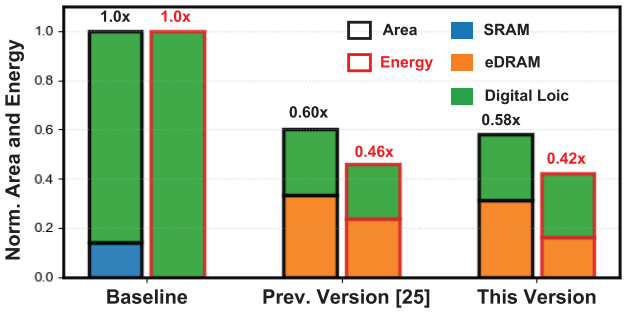


Fig. 8. Area and energy improvement of eLADs over the baseline digital CIM implementation.

from the other low-voltage computing circuits and the eDRAM array. Such isolation can mitigate the threshold voltage loss in write selectors of eDRAM cells, especially in low-voltage operating conditions, as shown in Section VI.

IV. ELAD TREE IMPLEMENTATIONS

A. Design Details of Optimized 3T eDRAM Cell

Before discussing the design tradeoffs of eLAD, we first elaborate on the implementation details and modifications of the 3 T eDRAM cell compared to our previous work as shown in Fig. 4 [25]. As outlined in Section II-B, the removal of additional capacitors at the storage node enables a more compact layout. To compensate for the resulting reduction in retention time without compromising area efficiency, we adopt an alternative approach to suppress the leakage. Rather than employing negative WWL voltages, which requires dedicated negative voltage generator [36], we modify WBL bias voltage from to VSS to VDD to naturally establish a negative V_{GS} for bit-"0."

Fig. 5 presents the Monte Carlo simulation results comparing leakage behavior under different configurations. When the WBL is biased at VSS, the bit-"1" rapidly decays to VSS within 3 μs at the fastest corner, thereby limiting the overall retention time of the eDRAM macro. In contrast, the leakage current can be greatly reduced for bit-"1" by biasing the WBL at VDD. Although this configuration causes bit-"0" to be charged toward bit-"1" gradually, the increased storage node voltage introduces a negative V_{GS}, which effectively suppresses the channel leakage current. To further mitigate the impact of increased bit-"0" voltage, the pair of read transistors in the eDRAM cell is changed from LVT to RVT devices. As a result, 220-mV voltage margin is preserved at 20 μs, leading to a more than 7$\times$ improvement in retention time, as presented in Section VI.

Additionally, the readout port design has been simplified to further reduce the area and energy consumption of the eDRAM array, as illustrated in the right column of Fig. 4. The previous design employed a C^2MOS-like dynamic latch to eliminate potential glitches during RBL precharging and discharging phases, at the cost of driving additional complementary control signals REN and RENB. The current implementation, similar to a TSPC register [37], employs a single NMOS transistor instead of a transmission gate while sharing the PRE signal. Although the modification introduces output glitches during RBL discharging, the total power and area are reduced by 8% and 3% due to simplified readout port design.

B. Efficient eLAD

Fig. 6(a) presents the implementation of the proposed LUT-based adder that utilizes compact 3 T eDRAM cells. The design incorporates a 16 $\times$ 10-bit eDRAM subarray with hierarchical write bitlines and the local RWL decoder shared by different CIM banks. In the CIM mode, the eLAD performs MAC operations of 4 $\times$ 1-bit inputs and 4 $\times$ 8-bit weights per cycle through LUT access. As demonstrated in Fig. 6(b), the high-density memory cells enable efficient LUT-based computing, with the implemented eDRAM cell occupying merely 0.145 μm^2, which is 11.9$\times$ smaller than that of a standard nine-track full adder cell.

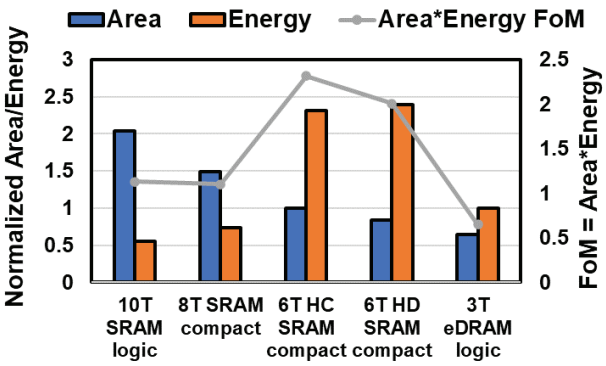
Fig. 6 illustrates the weight data mapping organized by corresponding activation binary codes. This configuration allows the input bits to be directly used as LUT access address, thereby making the decoder logic consist with normal memory operation except for all-zero inputs. Since such case produces zero outputs irrespective of stored weights, the corresponding wordline is power-gated in the CIM mode by turning off discharging. Although it appears to waste storage capacity, the additional wordline can both simplify the decoder logic in memory mode and serve as a scratchpad for in-memory encoding operation, as discussed in Section V.

Fig. 7 shows the detailed timing diagram of the eLAD and subsequent accumulation circuits for an 8-bit MAC operation. The entire process follows a three-stage pipeline: 1) eLAD partial sum generation; 2) adder tree accumulation; and 3) bit-shift accumulation. First, the input activations are fed to the CIM macro in a bit-serial, LSB-first manner. The precharge signal initializes the RBL to prepare for readout. Then, for every four bits of activation data, a corresponding RWL pulse is generated to retrieve the partial sum DOUT from the eDRAM LUT. Second, these partial sums are accumulated by the adder tree, with its output SUM sampled on the rising edge of the next precharge pulse. Finally, the complete MAC result RES is generated by the shift-adders over eight cycles.

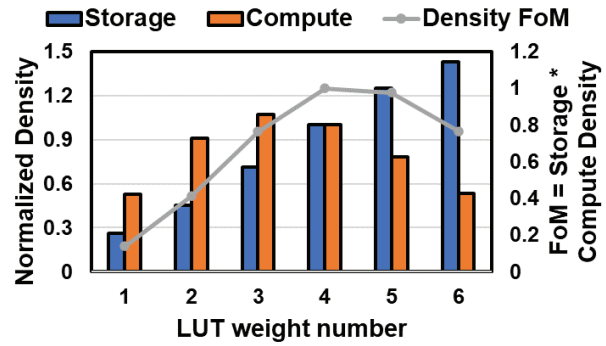
Compared to the baseline digital CIM composed of the standard digital cell and adopting the same input bit-serial architecture, as shown in Fig. 8, the proposed eLAD-based design achieves 40% area and 54% energy reduction while maintaining the same throughput and full precision. The improvements come from both the high-density memory cell and efficient LUT read operations. The current version achieves further reduction by simplifying the readout port, as illustrated in Fig. 4. In addition to the efficiency improvement, the proposed LUT-based computing strikes a balance of storage and computing. The eDRAM array dominates the overall area, leading to significant storage density in the memory mode.

Fig. 9 further explores the potential design space of LUT-based computing. First, we compare different LUT implementations using SRAM cells ranging from 10 to 6 T, which reveals a tradeoff between area and readout energy, as shown in Fig. 9(a). When accounting for dummy cells

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Fig. 9. Design space exploration of LUT-based adders on (a) memory cell and (b) weight number selection.

surrounding the 16 × 10-bit array, the 3 T eDRAM achieves a 36% and 22% area reduction compared to foundry-provided 6 T high-performance and high-density SRAM, respectively. Although the 10 T SRAM [38], featuring a precharging-free and single-ended readout port, achieves the highest energy efficiency, the additional transistors result in a 3× area overhead, making it impractical for area-sensitive LUT-based computing. By adopting the area-energy product as a figure of merit (FoM), the FoM curve demonstrates that the proposed eDRAM-based solution achieves the optimal tradeoff of area and energy efficiency.

In addition to memory cell selection, Fig. 9(b) analyzes the impact of different LUT depths. As expected, the number of weights results in an exponential growth in memory size. While increasing the storage ratio and memory density, the overall benefit of LUT-based computing diminishes gradually. Specifically, when >5 weights are contained, the area efficiency of eLAD drops below that of standard digital logic. By jointly considering storage and computational density, the FoM curve defined as their product reveals that the adopted four-weight LUT achieves the best tradeoff in density.

V. In-Memory Refreshing and Encoding Schemes

A. In-Memory Refreshing Implementations

After analyzing the advantages and tradeoffs of eDRAM-LUT-based CIM, we introduce how to overcome its refreshing

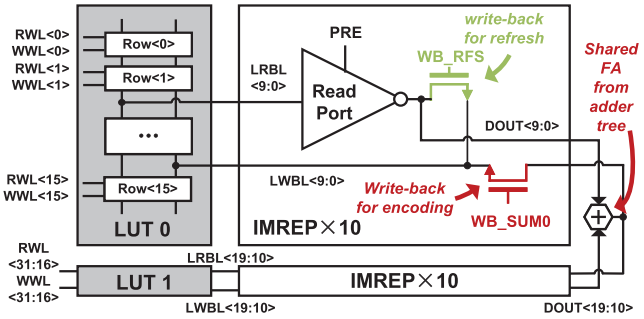


Fig. 10. Schematic of in-memory refreshing and encoding port.

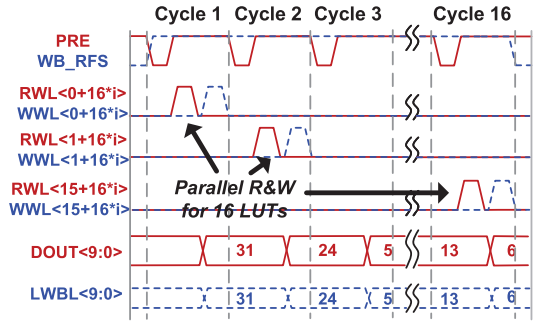


Fig. 11. Illustration of in-memory refreshing operations with detailed waveforms.

and programming overhead using the in-memory refreshing and encoding port. The proposed architecture contains a read port, which has been illustrated in Fig. 10, and two additional transistors marked in green and red, respectively.

The green transistor, controlled by the refresh write-back signal (WB_RFS), enables in situ refreshing through connecting the output of read port to LWBL. The red transistor, in contrast, redirects summation results from the following adder tree to LWBL for generating different weight combinations.

The detailed waveforms of in-memory refreshing are illustrated in Fig. 11. The timing of refreshing operations can be divided into three phases: precharging, discharging, and write-back. The WB_RFS signals are enabled following RWL pulse to realize write-back operation in one cycle. Unlike conventional row-by-row refreshing scheme, the in-memory refreshing activates different LUTs in parallel to achieve a 16× speedup. As shown in Section VI, the accelerated refreshing relaxes the constraint on eDRAM retention time.

B. In-Memory Encoding Implementations

The in-memory encoding operation is similar to the refreshing one with a main difference that different weight combinations are written back, instead of the original readout data. In order to minimize the area overhead of weight computing circuits, we reuse the adders from the subsequent adder tree. For the compatibility with the adder tree dataflow, where inputs are read from two neighboring LUTs, the initial weights require a double writing scheme.

The whole encoding process is illustrated in Fig. 12, which are divided into two stages: the weight loading and generation.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

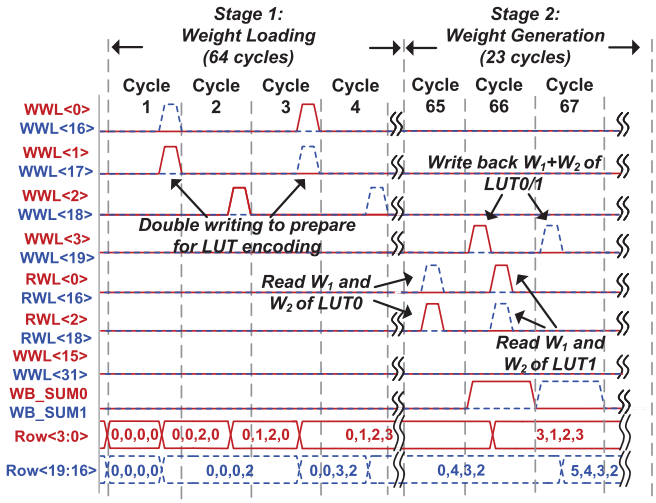


Fig. 12. Illustration of in-memory encoding operations with detailed waveforms.

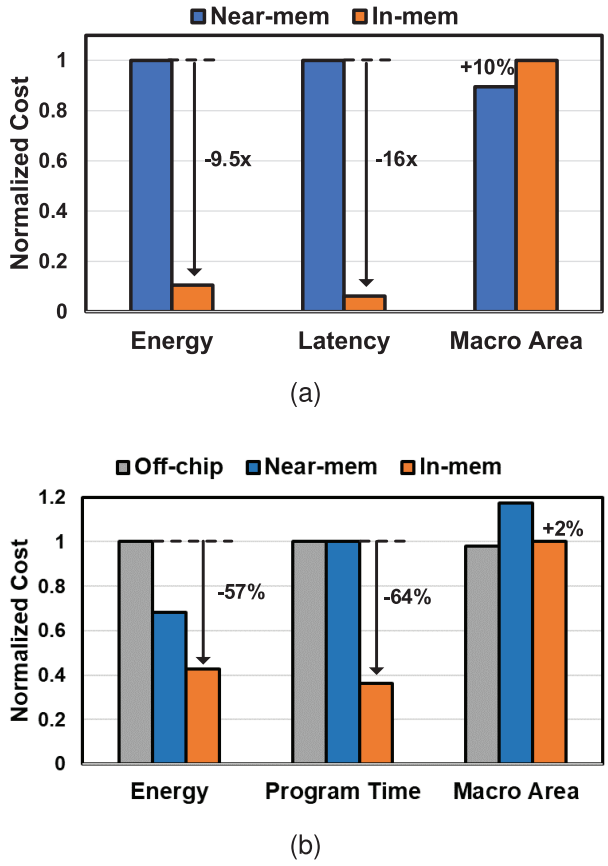


Fig. 13. Latency and efficiency improvements of in-memory (a) refreshing operations and (b) encoding operations.

Each weight data is stored in the corresponding LUT as well as its neighboring LUT to prepare for the next generation stage. The unused wordlines including WL0 serve as an intermediate weight buffer. Such double writing operation can be performed without latency overhead through activating both WWL in the same cycle.

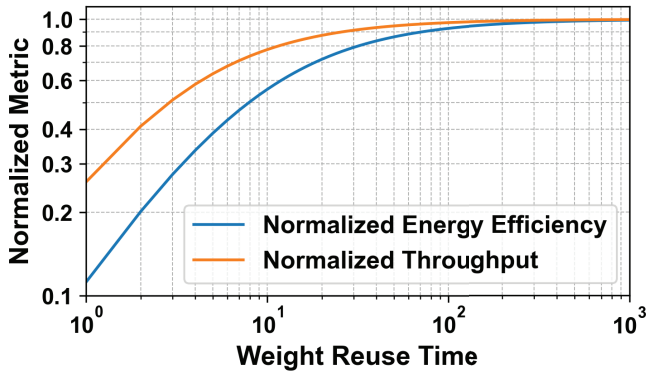


Fig. 14. Normalized macro energy efficiency and throughput considering the data loading time and energy under different weight reuse times.

During the generation process, the shared adder fetches intermediate data from two LUTs and write back the computation results by activating WB_SUM signal. To ensure sufficient setup time for weight computing, the write back signal, unlike the refreshing mode, is aligned with the precharging signal in the next cycle. Due to the parallel encoding, the weight generation stage takes up 23 cycles, 26% of the total programming time. Fig. 13 shows the whole process of in-memory encoding, where 1st, 7th, 11th, and 13th rows of each LUT are used to store the temporary value for LUT generation. By carefully arranging the order of weight read and write operations and leveraging the unused WL0, the whole process is data-collision free. Also, since the encoding stage requires no data interaction with external logic, it allows the LUT-based CIM macro to maintain a programming model similar to regular memory.

Fig. 14 summarizes the in-memory refreshing and encoding schemes by the evaluation results. Leveraging the parallel in-memory bandwidth, the proposed design achieves 16× and 2.9× reduction in refreshing and programming latency, respectively. In addition to the speed up, the in-memory technique improves 9.5× and 2.3× energy efficiency by reducing global data movement. The additional local read–write ports and write-back path from the adders increase the macro area by only 10% and 2%, respectively.

We also evaluate the normalized energy efficiency and throughput of the LUT-based CIM macro considering the programming overhead under different weight reuse times, as show in Fig. 15. Since the in-memory encoding stage consumes more power than normal computing due to concurrent read and write operations, it requires more computing cycles to amortize the energy overhead than that of throughput. Nevertheless, when the weight reuse exceeds 70×, which is typical in CNN models, the programming overhead accounts for less than 10%.

VI. Measurement Results

Fig. 16 presents the die photograph of the proposed eDRAM CIM macro and summarizes its main measurement results. The prototype chip is fabricated using 28-nm HKMG technology. The eDRAM macro occupies a 0.016 mm^2 area with a 40-Kb

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

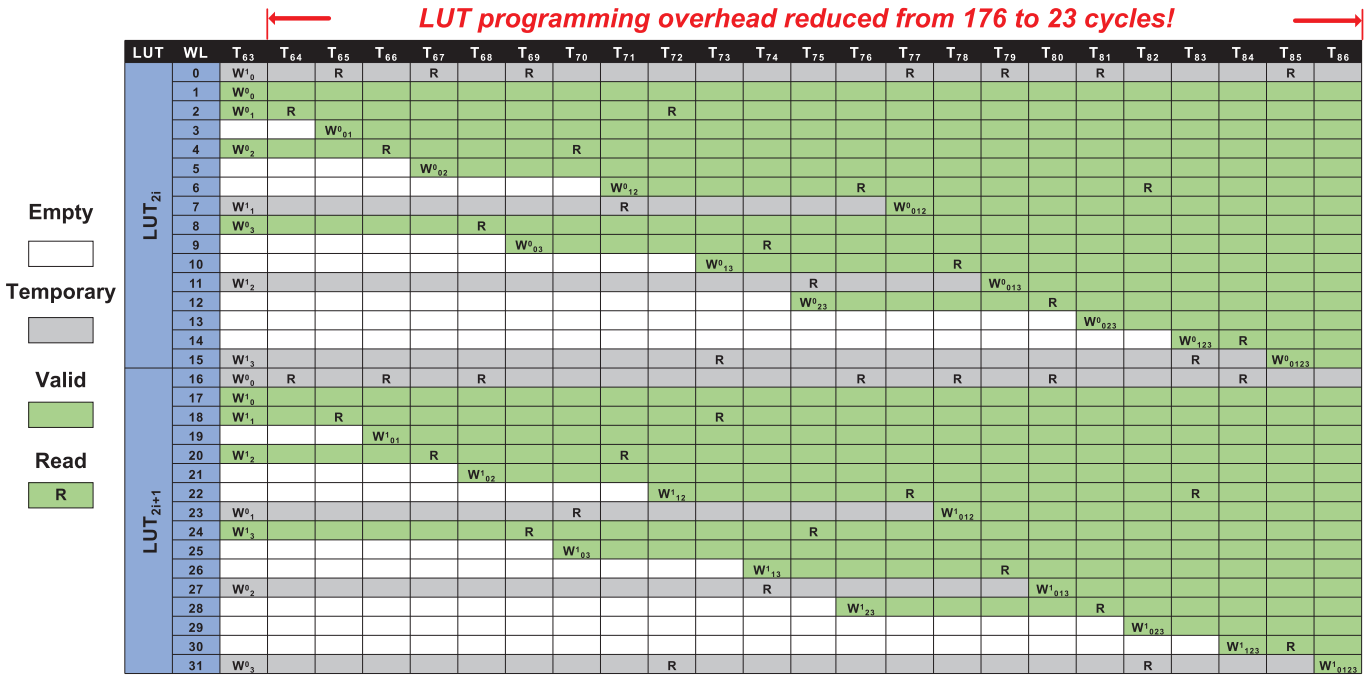
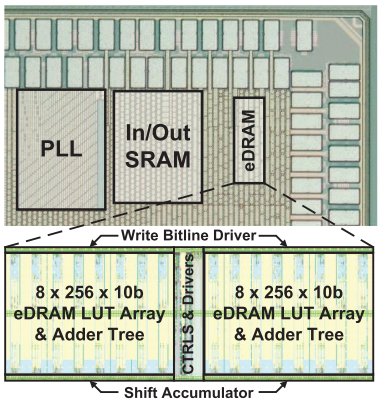


Fig. 15. Whole process of in-memory encoding for two neighboring LUTs in 23 cycles. The superscript of W denotes the number of the LUT, while the subscript denotes the combination of weights.



Chip Summary	
Technology	28nm HKMG
Bit Cell	3T eDRAM
Cell Area	0.145 um^2
Macro Area	0.016 mm^2
Capacity	40Kb
In/Out Channel	In:64/O:16
Supply Voltage	0.6-1.2V
Frequency	100-1050M
Storage Density	2.4Mb/mm^2
Peak Energy Efficiency	49.5 TOPS/W @ 8b, 0.6V
Peak Area Efficiency	16.5 TOPS/mm^2 @8b, 1.2V

Fig. 16. Die photograph of fabricated prototype and chip summary.

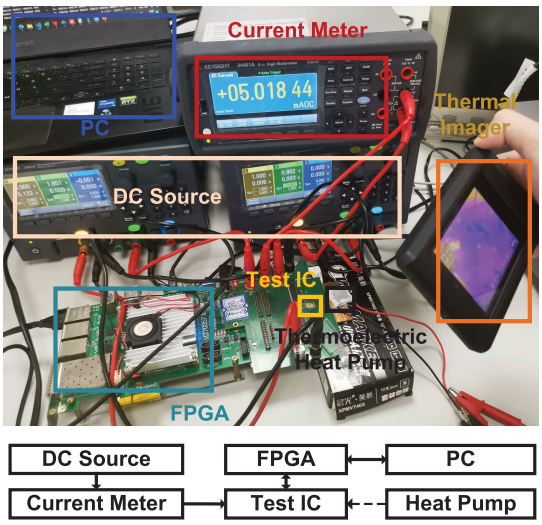


Fig. 17. Measurement setup.

array. The test chip also includes a PLL for peak frequency test and SRAM buffers for input and output data. Fig. 17 shows the testing setup, which includes the wire-bonded IC, the custom PCB, and an FPGA serving as a bridge between the chip and host PC. A thermoelectric heat pump controls the chip temperature, which is monitored by the thermal imager for retention test.

Fig. 18(a) demonstrates the voltage–frequency scaling characteristics. The separated voltage domains for write driver and computing circuits enable CIM operation at reduced voltages down to 0.6 V with 100-MHz frequency. The peak performance of 1.05 GHz is obtained at 1.2 V. Although the design exhibits minor performance degradation at higher voltages, compared to [25], due to limited discharging speed of RVT read transistors, its area efficiency is maintained due to optimized layout.

Fig. 18(b) presents the 8-bit energy-efficiency measurements across varying supply voltages and input bit sparsity levels, with weight bit sparsity fixed at 50%. A peak energy efficiency of 49.5 TOPS/W is achieved when operating at 0.6 V with 90% input sparsity.

Note that the voltages shown in Fig. 18 and subsequent Fig. 19 refer to the low-voltage domain of the computing circuits and the eDRAM array, which account for more than 80% of total power consumption, as indicated in Fig. 18. The high-voltage domain, which comprises the eDRAM write controllers and drivers, operates at 1.1 V when computing voltage is below 1.1 V and at 1.2 V otherwise.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

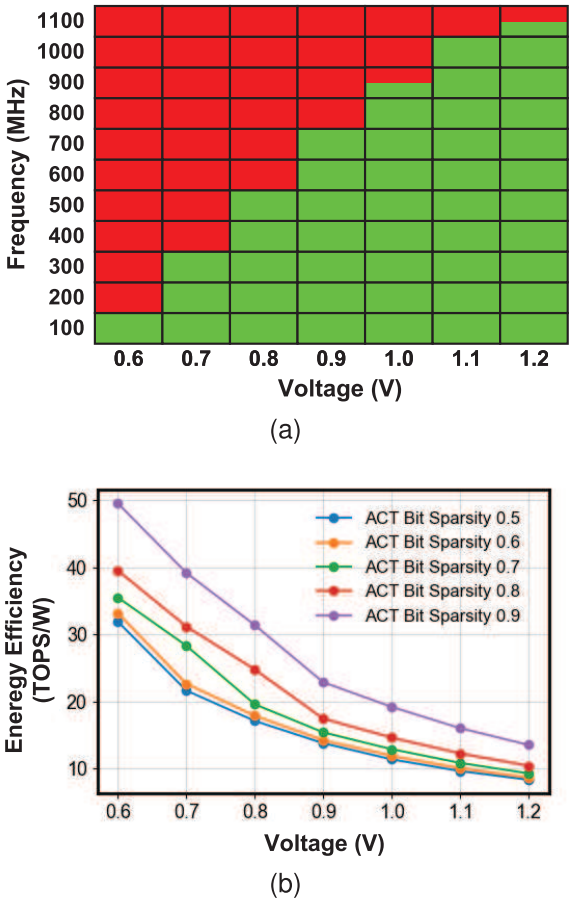


Fig. 18. (a) Shmoo plot and (b) voltage-efficiency scaling result.

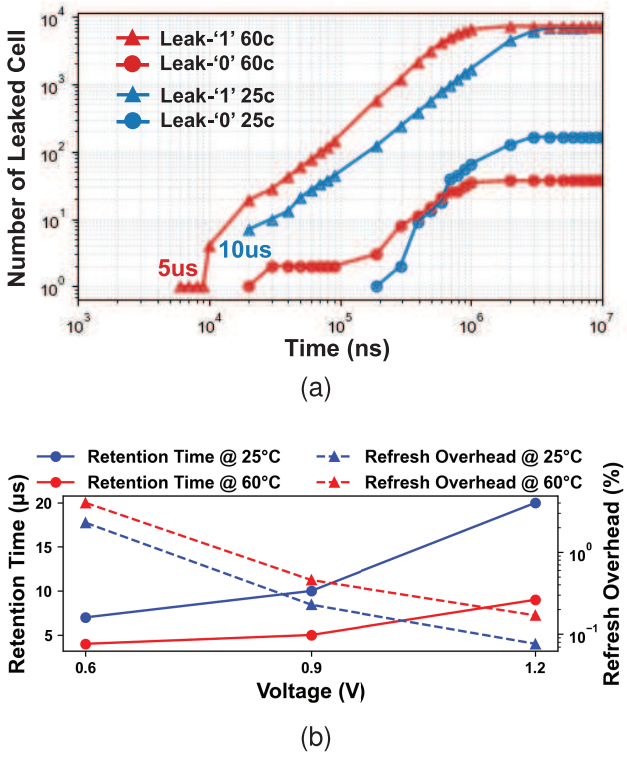


Fig. 19. (a) Number of leaked cells at 0.9-V voltage and 25 °C–60 °C and (b) retention time and refreshing overhead of different voltage and temperature corners.

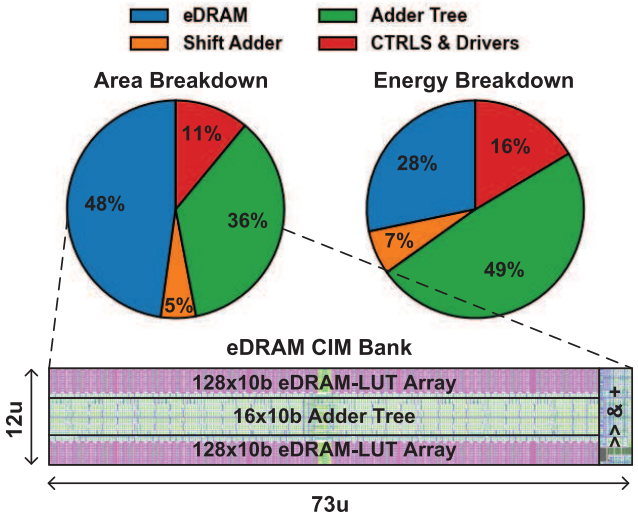


Fig. 20. Area and energy breakdown of the eDRAM-LUT-based CIM macro.

Fig. 19(a) presents the number of leaked eDRAM cell changing over time under varying temperatures at nominal voltage. The leaked cells have a different saturated number under various conditions due to our optimized bias scheme. As shown in the Monte Carlo simulation in Fig. 5, the final state of eDRAM cell lies between "1" and "0" as the result of balanced leakage paths to both WBL at VDD and substrate at VSS. For most cells, the balanced voltage is lower than the threshold of "1," causing them gradually decay to "0" over time. However, for some bits, the balanced states are still recognized at "1." Such balanced states are sensitive to PVT conditions, resulting in a varied distribution across different corners.

Fig. 19(b) further shows the retention time across different voltage and temperature corners. While the retention time is degraded to 4 μs at the worst case of 0.6 V and 60 °C, the eDRAM CIM macro maintains a refresh overhead less than 5%, due to the in-memory refresh technique.

Fig. 20 illustrates the detailed breakdown results. Compared to prior digital-dominated CIM designs, the proposed architecture demonstrates a more balanced computing-storage ratio. The bottom of the figure shows the layout of an eDRAM CIM bank employing a sandwich-like floorplan, where adder trees and shift-accumulation circuits occupy less than half area. For the energy breakdown, the adder tree accounts for 49% of the total power consumption, which is increased compared with our previous implementation [25], due to glitches at eDRAM-LUT outputs. However, the overall energy efficiency is improved due to simplified eDRAM readout circuits and control signals.

Table I compares this work with prior eDRAM- and SRAM-based analog and digital CIM designs. The proposed design combines the benefits of eDRAM and digital CIM approaches, offering both high density and accuracy. Compared to previous analog CIM designs, which suffered from the efficiency and

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON WITH PREVIOUS EDRAM AND SRAM CIM MACROS

	VLSI'23 C10-3	ISSCC'23 16.5	ISSCC'23 7.8	This Work	ISSCC'24 34.4	ISSCC'22 16.1	VLSI'22 C2.5
Technology	28nm	28nm	22nm	28nm	3nm	28nm	12nm
MAC Operation	Analog Charge	Analog Charge	Analog Charge	Digital Memory-LUT	Digital Adder tree	Digital Approx.	Digital Logic-LUT
Bit Cell	3T eDRAM	4T eDRAM	9T SRAM	3T eDRAM	6T SRAM	8T SRAM	6T SRAM
Macro Capacity	100Kb	100Kb	64Kb	8/40Kb (CIM/Mem)	60.75Kb	16Kb	8Kb
Macro Area (mm^2)	0.097	0.103	0.21	0.016	0.0157	0.033	0.0323
Storage Density (Mb/mm^2)	1.0	0.95	0.30	0.48/2.4 (CIM/Mem)	3.78	0.50	0.25
# of In/Out Ch	64 / 32	320 / 64	64 / 256	64 / 16	72 / 4	256 / 64	64 / 16
Input / Weight / Output Bits	I:8 W:9	I: 5/9 W: 4/8 O:13	I:8 W:8 O:22	I:1-8 W:8 O:22	I:12 W:12	I: 1 W: 1 O:7	I: 4-8 W: 4/8 O:18
8b* Energy Effi. (TOPS/W)	45.7 (@1.0V)	30.1 (@1V)	21.4/18.5 (@0.72/0.8V)	49.5/22.9/13.5 (@0.6/0.9/1.2V)	121 (@0.55V)	34.7/17.3/ 14.5 (@0.5/0.9/1.1V)	30.3 (@0.72V)
8b* Area Effi. ($TOPS/mm^2$)	0.81 (@1V)	2.03 (@1V)	1.19/1.44 (@0.72/0.8V)	1.57/11.0/16.5 (@0.6/0.9/1.2V)	66/123.8 (@0.55/0.9V)	4.3/9.5 (@0.9/1.1V)	10.4/14.3 (@0.72/0.92V)
Full Precision ?	No	No	Yes	Yes	Yes	No	Yes
Normalized** Density FoM***	0.81	1.92	0.36	39.6	6.8	4.75	0.88
Normalized** Efficiency FoM****	37.0 (@1V)	61.1 (@1V)	21.7 (@0.8V)	251.9 (@0.9V)	218.4 (@0.55V)	75.1 (@0.9V)	107.0 (@0.72V)

*Normalized to 8bit operations by IxW bits ** The area efficiency, energy efficiency and storage density are all normalized to 28nm based on the foundry SRAM bitcell size.
*** Density FoM = Peak Area Efficiency x Storage Density in memory mode **** Efficiency FoM = Energy x Area Efficiency @ Nominal voltage

TABLE II
NEURAL NETWORK PERFORMANCE AND EFFICIENCY BENCHMARKS

Neural Network Model	ResNet-50	ResNet-18
Dataset	ImagetNet-1k	
Bit Precision	8b/8b W/A	8b/8b W/A
Accuracy (%)	79.38	69.75
Compute/Programming Energy per Class (uJ)[1,2]	378.5/15.6	194.6/7.6
Throughput (fps)[1,2]	22.2	48.3

[1]Only consider macro programming, computing and refreshing energy/latency for batch size of 1 [2]Evaluated at 0.9V/700MHz

SNR tradeoff, this work achieves higher throughput and energy efficiency without compromising accuracy. Compared to other digital CIM implementations, including standard, approximate, and logic-LUT-based approaches, the proposed eDRAM-LUT-based design achieves the highest area and energy efficiency product at nominal voltage when normalized to 28-nm technology node. Besides, the proposed LUT-based design achieves a superior storage density of 2.4 Mb/mm^2 in the memory mode, leading to a 5.8× normalized density FoM. Although the effective density is reduced by 5× in CIM mode due to expanded LUT storage, it remains 1.9× higher than that of a 12-nm logic-LUT-based design.

Table II presents the benchmarking results of the proposed eDRAM-LUT-based macro on typical neural network models. It achieves 215.2/394.1 μJ per image energy consumption and 48.3/22.2 frames/s throughput for ResNet-18/ResNet-50 models on ImageNet dataset. Due to the efficient in-memory refreshing and encoding techniques, the total overhead of LUT programming and eDRAM refreshing is below 5%.

VII. CONCLUSION

Digital-based approach is a promising candidate to implement reliable and large-scale CIM architecture, due to its exceeding performance, robustness, and scalability. However, the existing digital CIM solutions mainly focus on optimizing computing logic and dataflow, instead of taking memory and computing circuits as a whole.

The proposed eDRAM-LUT-based design can break the limitation of digital logic and combine the benefits of high-density eDRAM cell and high-throughput digital CIM architecture. The power and area consuming digital datapath are replaced by efficient LUT. The eDRAM-LUT-based implementation achieves 42% and 58% area and energy reduction compared to traditional digital design methodology. To overcome the eDRAM leakage and LUT programming issues, the in-memory refreshing and weight encoding techniques are developed. Leveraging the parallel in-memory bandwidth, the refreshing and programming latency are reduced by 16× and 2.9×.

A 28-nm prototype is fabricated and measured to validate the proposed eDRAM-LUT-based CIM design. The measured results show a peak 8-bit energy and area efficiency of 49.5 TOPS/W and 16.5 $TOPS/mm^2$ with 2.4-Mb/mm^2 storage density in the memory mode. The efficiency FoM and density FoM surpass previous work by 1.2× and 5.8×.

REFERENCES

[1] Y.-H. Chen, T. Krishna, J. S. Emer, and V. Sze, "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 127–138, Jan. 2017.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.

[2] B. Moons, R. Uytterhoeven, W. Dehaene, and M. Verhelst, "Envision: A 0.26-to-10 TOPS/W subword-parallel dynamic-voltage-accuracy-frequency-scalable convolutional neural network processor in 28nm FDSOI," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2017, pp. 246–247.

[3] M. Horowitz, "Computing's energy problem (and what we can do about it)," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2014, pp. 10–14.

[4] R. Hameed et al., "Understanding sources of inefficiency in general-purpose chips," in *Proc. 37th Annu. Int. Symp. Comput. Archit.*, Jun. 2010, pp. 37–47.

[5] M. Kang, M.-S. Keel, N. R. Shanbhag, S. Eilert, and K. Curewitz, "An energy-efficient VLSI architecture for pattern recognition via deep embedding of computation in SRAM," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 8326–8330.

[6] P. Chi et al., "PRIME: A novel processing-in-memory architecture for neural network computation in ReRAM-based main memory," in *Proc. ACM/IEEE 43rd Annu. Int. Symp. Comput. Archit. (ISCA)*, Jun. 2016, pp. 27–39.

[7] H. Valavi, P. J. Ramadge, E. Nestler, and N. Verma, "A 64-tile 2.4-mb in-memory-computing CNN accelerator employing charge-domain compute," *IEEE J. Solid-State Circuits*, vol. 54, no. 6, pp. 1789–1799, Jun. 2019.

[8] W.-S. Khwa et al., "A 65nm 4Kb algorithm-dependent computing-in-memory SRAM unit-macro with 2.3ns and 55.8TOPS/W fully parallel product-sum operation for binary DNN edge processors," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 496–498.

[9] Q. Liu et al., "A fully integrated analog ReRAM based 78.4TOPS/W compute-in-memory chip with fully parallel MAC computing," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 500–502.

[10] H. Kim, Q. Chen, T. Yoo, T. T.-H. Kim, and B. Kim, "A 1–16b precision reconfigurable digital in-memory computing macro featuring column-MAC architecture and bit-serial computation," in *Proc. IEEE 45th Eur. Solid State Circuits Conf. (ESSCIRC)*, Sep. 2019, pp. 345–348.

[11] Y.-D. Chih et al., "An 89TOPS/W and 16.3TOPS/mm^2 all-digital SRAM-based full-precision compute-in memory macro in 22nm for machine-learning edge applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 252–254.

[12] X. Si et al., "A twin-8T SRAM computation-in-memory unit-macro for multibit CNN-based AI edge processors," in *Proc. IEEE J. Solid-State Circuits (JSSC)*, Apr. 2019, vol. 55, no. 1, pp. 189–202.

[13] J.-W. Su et al., "A 28nm 64Kb inference-training two-way transpose multibit 6T SRAM compute-in-memory macro for AI edge chips," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 240–242.

[14] X. Si et al., "A 28nm 64Kb 6T SRAM computing-in-memory macro with 8b MAC operation for AI edge chips," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 246–248.

[15] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm^2 fully-digital computing-in-memory macro supporting wide-range dynamic-voltage-frequency scaling and simultaneous MAC and write operations," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 65, Feb. 2022, pp. 1–3.

[16] H. Mori et al., "A 4nm 6163-TOPS/W/b 4790-TOPS/mm^2/b SRAM based digital-computing-in-memory macro supporting bit-width flexibility and simultaneous MAC and weight update," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 132–134.

[17] H. Fujiwara et al., "A 3nm, 32.5TOPS/W, 55.0TOPS/mm^2 and 3.78Mb/mm^2 fully-digital compute-in-memory macro supporting INT12 $\times$ INT12 with a parallel-MAC architecture and foundry 6T-SRAM bit cell," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 572–574.

[18] P. A. Hager et al., "Metis AIPU: A 12nm 15TOPS/W 209.6TOPS SoC for cost- and energy-efficient inference at the edge," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 212–214.

[19] M.-E. Shih et al., "NVE: A 3nm 23.2TOPS/W 12b-digital-CIM-based neural engine for high-resolution visual-quality enhancement on smart devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 360–362.

[20] S. Lee et al., "A 1ynm 1.25V 8Gb, 16Gb/s/pin GDDR6-based accelerator-in-memory supporting 1TFLOPS MAC operation and various activation functions for deep-learning applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 65, Apr. 2022, pp. 1–3.

[21] S. Lee et al., "Hardware architecture and software stack for PIM based on commercial DRAM technology: Industrial product," in *Proc. ACM/IEEE 48th Annu. Int. Symp. Comput. Archit. (ISCA)*, Jun. 2021, pp. 43–56.

[22] Y. He et al., "A 28nm 38-to-102-TOPS/W 8b multiply-less approximate digital SRAM compute-in-memory macro for neural-network inference," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 130–132.

[23] W. Sun et al., "A survey of computing-in-memory processor: From circuit to application," *IEEE Open J. Solid-State Circuits Soc.*, vol. 4, pp. 25–42, 2024.

[24] S. Kim et al., "DynaPlasia: An eDRAM in-memory-computing-based reconfigurable spatial accelerator with triple-mode cell for dynamic resource switching," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 256–258.

[25] Y. He et al., "A 28nm 2.4Mb/mm^2 6.9–16.3TOPS/mm^2 eDRAM-LUT-based digital-computing-in-memory macro with in-memory encoding and refreshing," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 578–580.

[26] D. Shin, J. Lee, J. Lee, and H.-J. Yoo, "14.2 DNPU: An 8.1TOPS/W reconfigurable CNN-RNN processor for general-purpose deep neural networks," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2017, pp. 240–241.

[27] J. Lee, C. Kim, S. Kang, D. Shin, S. Kim, and H.-J. Yoo, "UNPU: An energy-efficient deep neural network accelerator with fully variable weight bit precision," *IEEE J. Solid-State Circuits*, vol. 54, no. 1, pp. 173–185, Jan. 2019.

[28] C.-F. Lee et al., "A 12nm 121-TOPS/W 41.6-TOPS/mm^2 all digital full precision SRAM-based compute-in-memory with configurable bit-width for AI edge applications," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Jun. 2022, pp. 24–25.

[29] C. Tang, X. Lin, Z. Huang, W. Sun, H. Jia, and Y. Liu, "A 28nm 4.35TOPS/mm^2 transformer accelerator with basis-vector based ultra storage compression, decomposed computation and unified LUT-assisted cores," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Jun. 2024, pp. 1–2.

[30] S. Xie, C. Ni, A. Sayal, P. Jain, F. Hamzaoglu, and J. P. Kulkarni, "EDRAM-CIM: Compute-in-memory design with reconfigurable embedded-dynamic-memory array realizing adaptive data converters and charge-domain computing," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 248–250.

[31] K. Yoshioka, "A 818–4094 TOPS/W capacitor-reconfigured analog CIM for unified acceleration of CNNs and transformers," *IEEE J. Solid-State Circuits*, vol. 60, no. 5, pp. 1844–1855, May 2024.

[32] S. Zhao et al., "Be CIM or be memory: A dual-mode-aware DNN compiler for CIM accelerators," in *Proc. 30th ACM Int. Conf. Architectural Support Program. Lang. Operating Syst.*, Mar. 2025, pp. 63–78, doi: 10.1145/3676641.3716248.

[33] J. Song et al., "A calibration-free 15-level/cell eDRAM computing-in-memory macro with 3T1C current-programmed dynamic-cascoded MLC achieving 233-to-304-TOPS/W 4b MAC," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2023, pp. 1–2.

[34] B. Yan et al., "A 1.041-Mb/mm^2 27.38-TOPS/W signed-INT8 dynamic-logic-based ADC-less SRAM compute-in-memory macro in 28nm with reconfigurable bitwise operation for AI and embedded applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 65, Feb. 2022, pp. 188–190.

[35] C.-H. Lu, Y.-T. Hsu, B.-C. Wu, and T.-T. Liu, "A 270-mV 6T SRAM using row-based dual-phase VDD control in 28-nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 67, no. 12, pp. 4774–4783, Dec. 2020.

[36] A. Yigit, E. N. Casarrubias, R. Giterman, and A. Burg, "A 128-kbit GC-eDRAM with negative boosted bootstrap driver for 11.3$\times$ lower-refresh frequency at a 2.5% area overhead in 28-nm FD-SOI," *IEEE Solid-State Circuits Lett.*, vol. 6, pp. 13–16, 2023.

[37] B. Razavi, "TSPC logic [a circuit for all seasons]," *IEEE Solid StateCircuits Mag.*, vol. 8, no. 4, pp. 10–13, Fall. 2016.

[38] Y. Li et al., "A 28 nm 0.25–0.61 mW 31-60fps versatile SoC for diverse extreme edge ML workloads with flexible hetero-fabric dataflow orchestration and compute/storage-density-adjustable CIM," in *Proc. IEEE Eur. Solid-State Electron. Res. Conf. (ESSERC)*, Sep. 2024, pp. 149–152.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.



Yifan He (Member, IEEE) received the B.S. and Ph.D. degrees from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2019 and 2024, respectively.

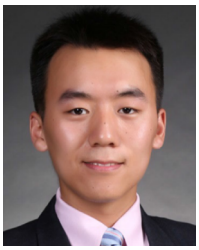
In March 2025, he works as a Post-Doctoral Researcher at the School of Electrical and Computer Engineering, Cornell Tech, New York, NY, USA. His research focuses on emerging nonvolatile memories and in-memory computing designs, with a focus on digital circuit technology for superior efficiency and compute density.

Dr. He was a recipient of the IEEE Solid-State Circuits Society (SSCS) Predoctoral Achievement Award from 2023 to 2024.



Shupei Fan (Graduate Student Member, IEEE) received the B.S. and M.S. degrees from Tsinghua University, Beijing, China, in 2019 and 2025, respectively.

His research interests include architecture designs on CNN and transformer accelerators.



Xuan Li (Graduate Student Member, IEEE) received the B.Eng. degree from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2015. He is currently pursuing the Ph.D. degree with Tsinghua University, Beijing, China.

He is the Founder of Wiener Technology, Beijing. His research interests include energy-efficient data converter design, mixed-signal and analog signal processing systems, and emerging sensing technology.



Luchang Lei (Graduate Student Member, IEEE) received the B.S. degree in electronic science and technology from Tsinghua University, Beijing, China, in 2022, where he is currently pursuing the Ph.D. degree in electronic science and technology with the Department of Electronic Engineering.

His research interests include in-memory processing architecture and circuit and hardware acceleration for homomorphic encryption.



Wenbin Jia (Graduate Student Member, IEEE) received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2022, where he is currently pursuing the Ph.D. degree.

His research interests include computing-in-memory circuit design and 3-D integration architectures.



Chen Tang (Member, IEEE) received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree.

His research interests include neural network compression and machine learning accelerator.



Yaolei Li (Graduate Student Member, IEEE) received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2021, where he is currently pursuing the Ph.D. degree.

His current research interests include computing-in-memory system design and 3-D integration architectures.



Zongle Huang (Graduate Student Member, IEEE) received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2023, where she is currently pursuing the Ph.D. degree.

Her research interests include distributed systems and architecture design for AI applications.



Zhike Du received the B.S. degree from the Department of Computer Science, Beihang University, Beijing, China, in 2005.

Since 2020, he has been working as a Hardware Engineer at Tsinghua University, Beijing. His research interests include computing-in-memory circuits, digital accelerator, and emerging AI applications.



Jinshan Yue (Member, IEEE) received the B.S. and Ph.D. degrees from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2016 and 2021, respectively.

From 2021 to 2024, he was a Post-Doctoral Researcher with the Institute of Microelectronics of the Chinese Academy of Sciences (IMECAS), Beijing, where he is currently an Associate Researcher. His current research interests include energy-efficient neural network processors, nonvolatile memory, and computing-in-memory systems.

Dr. Yue served as a TPC Member for IEEE A-SSCC and ICCAD.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.



Xueqing Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2007 and 2013, respectively.

He is currently an Associate Professor with the Department of Electronic Engineering, Tsinghua University. He has more than 170 publications and 30 China and U.S. patents. His research interests include mixed-signal circuit design, emerging memory, and memory-oriented computing exploration for AI acceleration.

Dr. Li is a Senior Member of CCF China. He received a few best paper awards, including HPCA'15, ASP-DAC'17, and 2016 IEEE TMSCS, several teaching and thesis awards, and the 2019 National Early-Career Award. He serves as a TPC Member for a few conferences, including ISSCC, DAC, ICCAD, DATE, and ASP-DAC. He is an Associate Editor of IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS, IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING (TETC), and *Journal of Electronics and Information Technology*, and a Guest Editor of a few journal issues, including IEEE JOURNAL OF SOLID-STATE CIRCUITS (JSSC), *International Journal of Circuit Theory and Applications*, and *Flexible and Printed Electronics*.



Huazhong Yang (Fellow, IEEE) received the B.S. degree in microelectronics and the M.S. and Ph.D. degrees in electronic engineering from Tsinghua University, Beijing, China, in 1989, 1993, and 1998, respectively.

In 1993, he joined the Department of Electronic Engineering, Tsinghua University, where he has been a Full Professor since 1998. He has been in charge of several projects, including projects sponsored by the National Science and Technology Major Project, the 863 Program, NSFC, and several international research projects. He has authored or co-authored over 500 technical articles, seven books, and over 180 granted Chinese patents. His research interests include wireless sensor networks, data converters, energy-harvesting circuits, nonvolatile processors, and brain-inspired computing.

Prof. Yang has served as a Navigating Committee Member for AsianHOST 2018 and a TPC Member for ASP-DAC 2005, APCCAS 2006, ICCCAS 2007, ASQED 2009, and ICGCS 2010. He received the Distinguished Young Researcher by NSFC in 2000, the Cheung Kong Scholar by Chinese Ministry of Education (CME) in 2012, the Science and Technology Award First Prize by China Highway and Transportation Society in 2016, the Technological Invention Award First Prize by CME in 2019, the Gold Prize of iNEA 2019, and several best paper awards, including ISVLSI 2012, FPGA 2017, NVMSA 2017, and ASP-DAC 2019. He has served as the Chair for the Northern China ACM SIGDA Chapter Science in 2014 and the General Co-Chair for ASP-DAC 2020.



Hongyang Jia (Member, IEEE) received the B.E. degree from Tsinghua University, Beijing, China, in 2014, and the M.A. and Ph.D. degrees from Princeton University, Princeton, NJ, USA, in 2016 and 2021, respectively.

From 2021 to 2022, he was a Post-Doctoral Researcher with NVIDIA, Santa Clara, CA, USA. Since August 2022, he has been with the Department of Electronic Engineering, Tsinghua University, where he is currently an Assistant Professor. His research focuses on advanced computing technologies and systems, rethinking how emerging computing fabrics can be integrated with architectures and algorithms to enhance system efficiency and performance. His primary research interests include approximate computing, mixed-signal computing, in-memory computing, energy-efficient circuits, and architectures for robotic computing and security applications.

Prof. Jia was a recipient or a co-recipient of the 2017 Outstanding Student Designer Award from ADI Inc., the 2022 Bede Liu Best Dissertation Award in electrical and computer engineering from Princeton ECE, and the 2024 Edison Patent Award. He serves on the technical program committees for ESSERC, DAC, and ICCAD Conferences.



Yongpan Liu (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees from Tsinghua University, Beijing, China, in 1999, 2002, and 2007, respectively.

He is currently a Full Professor (Cheung Kong Scholar) with the Department of Electronic Engineering, Tsinghua University.

Prof. Liu was a Program Committee Member for ISSCC, A-SSCC, and CICC. He has received the 40 Young Innovators Award DAC 2017, the Best Paper/Poster Award from ASPDAC in 2017 and 2021, the Micro Top Pick 2016, HPCA 2015, and the Design Contest Awards of ISLPED in 2012, 2013, and 2019. He served as the General Secretary for ASPDAC 2021 and the Technical Program Chair for ICAC2023. He was an Associate Editor of IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II, and *IET Cyber-Physical Systems*. He also served as A-SSCC2020/AICAS2022 Tutorial Speaker and an IEEE CASS Distinguished Lecturer in 2021.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:19:04 UTC from IEEE Xplore. Restrictions apply.